

EC 441 - Week 10 Solution Set

Transport Layer, UDP, and Reliable Data Transfer

Problem 1. Transport Layer Purpose

The network layer provides host-to-host delivery, while the transport layer provides process-to-process communication. This allows data arriving at one machine to be delivered to the correct application process.

Problem 2. Multiplexing and Demultiplexing

The transport layer uses port numbers and sockets. At the sender, multiplexing combines data from different applications and adds transport headers. At the receiver, demultiplexing reads the destination port and delivers the data to the correct socket.

Problem 3. UDP Header

The UDP header is 8 bytes. The UDP length field includes the UDP header plus the application data:

$$\text{UDP Length} = 8 \text{ bytes} + 250 \text{ bytes} = 258 \text{ bytes}$$

The UDP length field should contain 258 bytes.

Problem 4. UDP vs. TCP

One example is DNS. DNS often uses UDP because it usually involves a small request and a small response. Avoiding a TCP handshake makes the lookup faster.

Another example is VoIP or real-time gaming. These applications care more about receiving fresh data quickly than retransmitting old data. A late voice packet or late game update may no longer be useful.

Problem 5. Transmission Time

First convert the packet size from bytes to bits:

$$1500 \text{ bytes} \times 8 = 12000 \text{ bits}$$

Use the transmission time formula:

$$t_{tx} = L / B$$

Substitute:

$$t_{tx} = 12000 / 100000000 = 0.00012 \text{ seconds}$$

Convert to microseconds:

$$0.00012 \text{ seconds} \times 1000000 = 120 \text{ microseconds}$$

Final answer: 0.00012 seconds = 120 microseconds.

Problem 6. Stop-and-Wait Utilization

Given:

$$B = 100000000 \text{ bits/s, } RTT = 0.08 \text{ seconds, } L = 12000 \text{ bits}$$

First calculate transmission time:

$$t_{tx} = 12000 / 100000000 = 0.00012 \text{ seconds}$$

Use the Stop-and-Wait utilization formula:

$$U = t_{tx} / (RTT + t_{tx})$$

Substitute:

$$U = 0.00012 / (0.08 + 0.00012) = 0.00012 / 0.08012 \approx 0.001497$$

As a percentage:

$$0.001497 \times 100 \approx 0.1497\%$$

Final answer: $U \approx 0.001497$, or about 0.15%.

Problem 7. Useful Throughput

Use:

$$\text{Throughput} = U \times B$$

Substitute:

$$\text{Throughput} = 0.001497 \times 100000000 \approx 149700 \text{ bits/s}$$

Final answer: about 149.7 kb/s. Even though the link bandwidth is 100 Mb/s, Stop-and-Wait wastes most of the link because it waits for an ACK after every packet.

Problem 8. Bandwidth-Delay Product

Given:

$$B = 500000000 \text{ bits/s, } RTT = 0.1 \text{ seconds}$$

Use:

$$BDP = B \times RTT$$

Substitute:

$$BDP = 500000000 \times 0.1 = 50000000 \text{ bits}$$

Convert to megabits:

$$50000000 \text{ bits} = 50 \text{ Mb}$$

Final answer: $BDP = 50000000 \text{ bits} = 50 \text{ Mb}$.

Problem 9. Required Window Size

From Problem 8:

$$BDP = 50000000 \text{ bits}$$

Convert packet size:

$$1000 \text{ bytes} \times 8 = 8000 \text{ bits}$$

Use:

$$W = \text{BDP} / \text{packet size}$$

Substitute:

$$W = 50000000 / 8000 = 6250 \text{ packets}$$

Final answer: 6250 packets. The sender would need about 6250 packets in flight to fully utilize the link.

Problem 10. Why Sequence Numbers Matter

Sequence numbers are necessary because the receiver must know whether the retransmitted packet is new data or a duplicate. If the ACK is lost, the sender retransmits the same packet. Without a sequence number, the receiver might deliver the same data to the application twice. With a sequence number, the receiver can recognize it as a duplicate, discard the data, and resend the ACK.

Problem 11. Go-Back-N Packet Loss

The sender sends packets 0, 1, 2, and 3. Packet 1 is lost. The receiver gets packet 0 and accepts it. Then it receives packets 2 and 3, but those packets are out of order because the receiver is still expecting packet 1.

In Go-Back-N, the receiver discards out-of-order packets. Therefore, packets 2 and 3 are discarded.

When the timer expires, the sender retransmits packet 1 and all later packets in the current window.

Final answer: the receiver discards packets 2 and 3. The sender retransmits packets 1, 2, and 3.

Problem 12. Selective Repeat Packet Loss

The sender sends packets 0, 1, 2, and 3. Packet 1 is lost. The receiver gets packet 0 and accepts it. Then it receives packets 2 and 3. In Selective Repeat, the receiver can buffer out-of-order packets if they are inside the receive window.

So the receiver buffers packets 2 and 3 and sends individual ACKs for them. When the timer for packet 1 expires, the sender retransmits only packet 1.

Final answer: the receiver buffers packets 2 and 3. The sender retransmits only packet 1. After packet 1 arrives, the receiver can deliver packets 1, 2, and 3 in order.

Problem 13. Sequence Number Constraint for Go-Back-N

Given $k = 4$, Go-Back-N uses:

$$N \leq 2^k - 1$$

Substitute:

$$N \leq 2^4 - 1 = 16 - 1 = 15$$

Final answer: $N_{\max} = 15$.

Problem 14. Sequence Number Constraint for Selective Repeat

Given $k = 4$, Selective Repeat uses:

$$N \leq 2^{(k - 1)}$$

Substitute:

$$N \leq 2^{(4 - 1)} = 2^3 = 8$$

Final answer: $N_{\max} = 8$.

Problem 15. TCP Connection to GBN and SR

TCP takes cumulative ACKs from the Go-Back-N style of reliable transfer. A TCP ACK confirms that all bytes up to a certain point have been received.

TCP also uses receiver buffering, which is more like Selective Repeat. If later segments arrive before a missing earlier segment, TCP can hold those later segments until the gap is filled.

With the SACK option, TCP becomes even more similar to Selective Repeat because the receiver can tell the sender which byte ranges have already arrived.